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Storage apparatus, storage method, and computer-readable storage medium

Abstract

A storage apparatus includes a biological sensor configured to detect biological information on a user, an image acquisition unit configured to acquire, from an image capturing unit, an image that is captured around the user, an image processing unit configured to separate the image around the user into a line-of-sight direction image related to an image in a line-of-sight direction of the user and a peripheral image related to an image other than the line-of-sight direction image, an encryption unit configured to generate an encryption key for encrypting the line-of-sight direction image based on the biological information on the user, and encrypts the line-of-sight direction image by using the encryption key, and a storage control unit configured to store the line-of-sight direction image that is encrypted by the encryption unit in a storage unit.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a Continuation of International Application No. PCT/JP2021/034825, filed Sep. 22, 2021, which designates the United States, incorporated herein by reference, and which claims the benefit of priority from Japanese Patent Applications No. 2020-159527, No. 2020-160129, and No. 2020-160130, each filed Sep. 24, 2020, all of which are incorporated herein by reference.

BACKGROUND

(1) The present disclosure relates to a storage apparatus, a storage method, and a computer-readable

storage medium.

(2) As wearable devices that are worn by users, devices, such as a head mounted display (EMD) type and an eyeglass type, that are worn in a wide variety of manners have come out. The wearable devices as described above include a device that has a function to capture a moving image in a line-of-sight direction of a user by a camera, and record a captured video.

(3) For example, Japanese Laid-open Patent Publication No. 2003-184351 describes a technology for matching a place that is viewed by a photographer and a place that is being captured in an image capturing device that is worn on an ear or a head. Japanese Laid-open Patent Publication No. 2000-341632 describes an apparatus that has a function to prevent snooping and falsification of image data that is digitally recorded.

(4) Here, there is a need to appropriately store an image in a line-of-sight direction of a user among images that are captured by a wearable device that is worn by the user, in such a manner that the image is not easily accessible in order to protect privacy of the user and privacy of a target.

SUMMARY

(5) A storage apparatus according to an aspect of the present disclosure includes: a biological sensor configured to detect biological information on a user; an image acquisition unit configured to acquire, from an image capturing unit, an image that is captured around the user; an image processing unit configured to separate the image around the user into a line-of-sight direction image related to an image in a line-of-sight direction of the user and a peripheral image related to an image other than the line-of-sight direction image; an encryption unit configured to generate an encryption key for encrypting the line-of-sight direction image based on the biological information on the user, and to encrypt the line-of-sight direction image by using the encryption key; and a storage control unit configured to store the line-of-sight direction image that is encrypted by the encryption unit in a storage unit.

(6) A storage method according to an aspect of the present disclosure includes the steps of: detecting biological information on a user; acquiring, from an image capturing unit, an image that is captured around the user; separating the image around the user into a line-of-sight direction image related to an image in a line-of-sight direction of the user and a peripheral image related to an image other than the line-of-sight direction image; generating an encryption key for encrypting the line-of-sight direction image based on the biological information on the user; encrypting the line-of-sight direction image by using the encryption key; and storing the encrypted line-of-sight direction image in a storage unit.

(7) A non-transitory computer-readable storage medium according to an aspect of the present disclosure stores a program causing a computer to execute: detecting biological information on a user; acquiring, from an image capturing unit, an image that is captured around the user; separating the image around the user into a line-of-sight direction image related to an image in a line-of-sight direction of the user and a peripheral image related to an image other than the line-of-sight direction image; generating an encryption key for encrypting the line-of-sight direction image based on the biological information on the user; encrypting the line-of-sight direction image by using the encryption key; and storing the encrypted line-of-sight direction image in a storage unit.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1A is a schematic diagram illustrating a first example of a storage apparatus according to a first embodiment.

(2) FIG. 1B is a schematic diagram illustrating the first example of the storage apparatus according to the first embodiment.

(3) FIG. 2 is a schematic diagram illustrating a second example of the storage apparatus according

to the first embodiment.

(4) FIG. 3 is a block diagram illustrating a configuration example of the storage apparatus according to the first embodiment.

(5) FIG. 4 is a flowchart illustrating an example of the flow of a process performed by the storage apparatus according to the first embodiment.

(6) FIG. 5 is a flowchart illustrating an example of the flow of a process performed by a storage apparatus according to a second embodiment.

(7) FIG. 6 is a block diagram illustrating a configuration example of a storage apparatus according to a third embodiment.

(8) FIG. 7A is a diagram for explaining encryption in an ECB mode.

(9) FIG. 7B is a diagram for explaining decryption in the ECB mode.

(10) FIG. 8A is a diagram for explaining encryption in an OFB mode.

(11) FIG. 8B is a diagram for explaining decryption in the OFB mode.

(12) FIG. 9 is a flowchart illustrating the flow of a process performed by the storage apparatus according to the third embodiment.

(13) FIG. 10 is a diagram for explaining a method of determining whether the storage apparatus has moved a predetermined distance or more.

(14) FIG. 11 is a diagram illustrating a configuration example of a storage apparatus according to a fourth embodiment.

(15) FIG. 12 is a flowchart illustrating an example of the flow of a process performed by the storage apparatus according to the fourth embodiment.

DETAILED DESCRIPTION

(16) Embodiments according to the present disclosure will be described in detail below with reference to the accompanying drawings. The present disclosure is not limited by the embodiments below, and when a plurality of embodiments are provided, the present disclosure includes a combination of the embodiments. In addition, in the embodiments below, the same components are denoted by the same reference symbols, and repeated explanation will be omitted.

First Embodiment

(17) FIG. 1A and FIG. 1B are schematic diagrams illustrating a first example of a storage apparatus according to a first embodiment. FIG. 1A illustrates a front side of a user U, and FIG. 1B illustrates a back side of the user U. As illustrated in FIG. 1A and FIG. 1B, a storage apparatus 10 is a wearable device that is worn on a body of the user U. As illustrated in FIG. 1A and FIG. 1B, the storage apparatus 10 includes a device 10A that is worn on eyes of the user U, a device 10B that is arranged on the device 10A, a device 10C.sub.L that is worn on a left ear of the user U, a device 10C.sub.R that is worn on a right ear of the user U, and a device 10D that is worn on the back of a head of the user U. The device 10A is an eyeglass-type display unit that displays a video. The device 10B is a camera that captures a video in a line-of-sight direction of the user U. The device 10C.sub.L is a microphone that collects sound from a left direction of the user U. The device 10C.sub.R is a microphone that collects sound from a right direction of the user U. The device 10C.sub.L and the device 10C.sub.R may be binaural microphones. The device 10D is a camera that captures a video behind the user U.

(18) FIG. 2 is a schematic diagram illustrating a second example of the storage apparatus according to the first embodiment. FIG. 2 illustrates the front side of the user U. As illustrated in FIG. 2, the storage apparatus 10 includes a device 10E that is worn on the eyes of the user U, the device 10B that is arranged on the device 10D, the device 10C.sub.L that is worn on the left ear of the user U, the device 10C.sub.R that is worn on the right ear of the user U, and a device 10F that is worn on a top of the head of the user U. The device 10D is a goggle-type display unit that displays a video, and is what is called an HMD. The device 10F is a camera with a fisheye lens that captures a 360-degree image around the user U. The device 10F may be what is called an omnidirectional camera.

(19) Storage Apparatus

(20) A configuration example of the storage apparatus according to the first embodiment will be described with reference to FIG. 3. FIG. 3 is a block diagram illustrating a configuration example of the storage apparatus according to the first embodiment.

(21) As illustrated in FIG. 3, the storage apparatus **10** includes a microphone **12**, a camera **14**, a biological sensor **16**, an input unit **18**, an output unit **20**, a communication unit **22**, a storage unit **24**, and a control unit **26**.

(22) The microphone **12** is a microphone that detects sound (sound wave information) around the storage apparatus **10** (the user U). The microphone **12** includes, for example, a left microphone **12A** and a right microphone **12B**.

(23) The left microphone **12A** is worn on, for example, the left ear of the user U. The left microphone **12A** detects sound in a left direction of the storage apparatus **10** (the user U). The right microphone **12B** is worn on, for example, the right ear of the user U. The right microphone **12B** detects sound in a right direction of the storage apparatus **10** (the user U). The left microphone **12A** and the right microphone **12B** may be, for example, binaural microphones. Meanwhile, the microphones included in the microphone **12** are not limited to the left microphone **12A** and the right microphone **12B**. The positions and the number of the microphones **12** that are arranged in the storage apparatus **10** may be arbitrary. For example, with use of the microphone **12** and a directional microphone with directivity, it is possible to store sound on the front, back, left, and right around the storage apparatus **10** (the user U) in a multiplexed manner.

(24) The camera **14** is an image capturing device, and captures an image around the storage apparatus **10** by detecting visible light around the storage apparatus **10** (the user U). The camera **14** may be a video camera that captures an image at a predetermined frame rate. The camera **14** includes a line-of-sight direction camera **14A** and a peripheral camera **14B**.

(25) The line-of-sight direction camera **14A** captures an image in the line-of-sight direction of the user U. The line-of-sight direction camera **14A** is arranged in a display unit **20A** of the output unit **20** so as to be able to capture an image in the line-of-sight direction of the user U, for example. The line-of-sight direction camera **14A** may be arranged in a portion other than the display unit **20A** of the output unit **20**.

(26) The peripheral camera **14B** captures an image in a range other than the line-of-sight direction of the user U around the storage apparatus **10** (the user U). The peripheral camera **14B** is arranged on the back of the head of the user U and captures an image on the back side of the user U, for example.

(27) The positions and the number of the line-of-sight direction camera **14A** and the peripheral camera **14B** arranged in the storage apparatus **10** may be arbitrary. The line-of-sight direction camera **14A** may be configured in an integrated manner with the peripheral camera **14B**. In this case, the camera **14** may be an omnidirectional camera that is worn on the top of the head of the user U. The omnidirectional camera captures an image of a range of 360 degrees around the storage apparatus **10** (the user U). The camera **14** may include a plurality of cameras and the plurality of cameras may capture images in the range of 360 degrees around the storage apparatus **10** (the user U). The camera **14** may include, for example, a gimbal mechanism, and may capture an image in the range of 360 degrees around the storage apparatus **10** (the user U) while being rotated by the gimbal mechanism.

(28) The biological sensor **16** is a sensor that detects biological information on the user U. The biological sensor **16** may be arranged at an arbitrary position as long as it is possible to detect the biological information on the user U. The biological information acquired by the biological sensor **16** may be universal information that is unique to the user U. In the present embodiment, the biological information may be, for example, a fingerprint, an iris pattern, a vein shape of a finger and a hand, and the like, but is not limited thereto. The biological information may be a combination of information on a fingerprint or the like and information on a feature of a face. The biological information may be genetic information on the user U. The biological sensor **16** is

implemented by, for example, a fingerprint sensor, a vein sensor, a camera, or the like.

(29) The input unit **18** receives various kinds of operation on the storage apparatus **10**. The input unit **18** is implemented by, for example, a button, a touch panel, and the like.

(30) The output unit **20** outputs various kinds of information. The output unit **20** includes, for example, the display unit **20A** and a sound output unit **20B**. The display unit **20A** displays various kinds of videos. In the present embodiment, the display unit **20A** is, for example, an eyeglasses-type display or an HMD. The sound output unit **20B** outputs various kinds of sound. The sound output unit **20B** is a speaker.

(31) The communication unit **22** performs communication with a different external apparatus. The communication unit **22** is implemented by, for example, a communication module, such as Wi-Fi (registered trademark) or Bluetooth (registered trademark). The communication unit **22** may have a function to perform communication with a different external apparatus in a wired manner, for example.

(32) The storage unit **24** is a memory for storing various kinds of information, such as details of calculation performed by the control unit **26** and programs, and includes at least one of a main storage device, such as a random access memory (RAM) or a read only memory (ROM), and an external storage apparatus, such as a hard disk drive (HDD), for example.

(33) The storage unit **24** stores therein a learning model **24A** and biological information **24B**. The learning model **24A** is an AI model that is used for recognizing a specific target that is included in an image, on the basis of the image. The learning model **24A** may be, for example, a learning model using a convolutional neural network (CNN). The learning model **24A** is used for recognizing, for example, privacy information related to privacy, such as a face of a person, a license plate of a vehicle, or a house nameplate. The learning model **24A** may be learned so as to be able to recognize a copyrighted material that is protected by copyright, for example. The biological information **24B** is biological information that includes, for example, a fingerprint, an iris pattern, or a vein shape of a finger and a hand of the user U. In other words, the biological information **24B** is biological information on a person who is permitted to use the storage apparatus **10**.

(34) The control unit **26** controls operation of each of the units of the storage apparatus **10**. The control unit **26** is implemented by, for example, causing a central processing unit (CPU), a micro processing unit (MPU), or the like to execute a program stored in a storage unit (not illustrated) by using a RAM or the like as a work area. The control unit **26** may be implemented by an integrated circuit, such as an application specific integrated circuit (ASIC) or a field programmable gate arrays (FPGA), for example. The control unit **26** may be implemented by a combination of a hardware and software.

(35) The control unit **26** includes a sound acquisition unit **30**, an image acquisition unit **32**, a biological information acquisition unit **34**, an authentication unit **36**, a sound processing unit **38**, an image processing unit **40**, an image recognition unit **42**, an image editing unit **44**, a multiplexing unit **46**, an encryption unit **48**, a storage control unit **50**, an output control unit **52**, and a communication control unit **54**.

(36) The sound acquisition unit **30** causes the microphone **12** to detect sound around the storage apparatus **10** (the user U). The sound acquisition unit **30** acquires the sound that is detected by the microphone **12**. Details of a process performed by the sound acquisition unit **30** will be described later.

(37) The image acquisition unit **32** causes the camera **14** to capture an image around the storage apparatus **10** (the user U). The image acquisition unit **32** acquires the image that is captured by the camera **14**. Details of a process performed by the image acquisition unit **32** will be described later.

(38) The biological information acquisition unit **34** causes the biological sensor **16** to detect biological information on the storage apparatus **10** (the user U). The biological information acquisition unit **34** acquires the biological information that is detected by the biological sensor **16**. Details of a process performed by the biological information acquisition unit **34** will be described

later.

(39) The authentication unit **36** authenticates the user U. The authentication unit **36** performs authentication on whether to allow the user U to use the storage apparatus **10**, on the basis of the biological information that is acquired by the biological information acquisition unit **34**. Details of a process performed by the authentication unit **36** will be described later.

(40) The sound processing unit **38** performs various kinds of sound signal processing on the sound that the sound acquisition unit **30** has acquired from the microphone **12**. Details of a process performed by the sound processing unit **38** will be described later.

(41) The image processing unit **40** performs various kinds of image processing on the image that the image acquisition unit **32** has acquired from the camera **14**. Details of a process performed by the image processing unit **40** will be described later.

(42) The image recognition unit **42** recognizes a specific target from the image that is acquired by the image acquisition unit **32**. Details of a process performed by the image recognition unit **42** will be described later.

(43) The image editing unit **44** performs various kinds of editing processes on the image that is acquired by the image acquisition unit **32**. The image editing unit **44** performs an editing process on the specific target that is recognized by the image recognition unit **42**. A process performed by the image editing unit **44** will be described later.

(44) The multiplexing unit **46** multiplexes the sound that is acquired by the sound acquisition unit **30** and the sound that is acquired by the image acquisition unit **32**. A multiplexing method may be arbitrary. Details of a process performed by the multiplexing unit **46** will be described later.

(45) The encryption unit **48** encrypts data that is multiplexed by the multiplexing unit **46**. The encryption unit **48** encrypts the data that is multiplexed by the multiplexing unit **46**, by using a predetermined encryption method and an encryption key. Details of a process performed by the encryption unit **48** will be described later.

(46) The storage control unit **50** stores various kinds of information in the storage unit **24**. Details of a process performed by the storage control unit **50** will be described later.

(47) The output control unit **52** causes the output unit **20** to output various kinds of information. The output control unit **52** causes the display unit **20A** to display various kinds of videos. The output control unit **52** causes the sound output unit **20B** to output various kinds of sound.

(48) The communication control unit **54** controls the communication unit **22**. The communication control unit **54** causes the communication unit **22** to transmit and receive various kinds of information to and from an external apparatus.

(49) Process Performed by Storage Apparatus

(50) A process performed by the storage apparatus according to the first embodiment will be described with reference to FIG. 4. FIG. 4 is a flowchart illustrating an example of the flow of a process performed by the storage apparatus according to the first embodiment.

(51) The control unit **26** acquires sound information on sound around the storage apparatus **10** (the user U) (Step S10). Specifically, the sound acquisition unit **30** acquires sound in the left direction of the storage apparatus **10** (the user U), which is detected by the left microphone **12A**. The sound acquisition unit **30** acquires sound in the right direction of the storage apparatus **10** (the user U), which is detected by the right microphone **12B**. Then, the process goes to Step S12.

(52) The control unit **26** performs a sound compression process on the sound information that is acquired by the sound acquisition unit **30** (Step S12). Specifically, the sound processing unit **38** performs sound compression signal processing on the sound that the sound acquisition unit **30** has acquired from the left microphone **12A** and the sound that the sound acquisition unit **30** has acquired from the right microphone **12B**, and generates a stereo signal in an MP3 format or the like, for example. Then, the process goes to Step S14.

(53) The control unit **26** acquires image information on an image around the storage apparatus **10** (the user U) (Step S14). Specifically, the image acquisition unit **32** acquires a line-of-sight direction

image related to the image in the line-of-sight direction of the user U, which is captured by the line-of-sight direction camera **14A**. The image acquisition unit **32** acquires a peripheral image related to the image in a peripheral area other than the line-of-sight direction of the user U, which is captured by the peripheral camera **14B**. Then, the process goes to Step **S16**.

(54) The control unit **26** determines whether a processing target image is the line-of-sight direction image (Step **S16**). Specifically, to separately perform image processing on the line-of-sight image direction and the peripheral direction image, the image processing unit **40** separates the image that is acquired by the image acquisition unit **32** into the line-of-sight image direction and the peripheral direction image. If it is determined that the image is the line-of-sight direction image (Step **S16**; Yes), the process goes to Step **S18**. If it is determined that the image is the peripheral image (Step **S16**; No), the process goes to Step **S32**. In other words, in the present embodiment, if it is determined as Yes at Step **S16**, the image processing is performed on the line-of-sight image direction, and, if it is determined as No at Step **S16**, the image processing is performed on the peripheral image.

(55) If it is determined as Yes at Step **S16**, the control unit **26** performs an image compression process on the line-of-sight direction image (Step **S18**). Specifically, the image processing unit **40** compresses the line-of-sight direction image by codec of an arbitrary system, such as Moving Picture Experts Group (MPEG)-4 or H.264, and converts the image into an arbitrary file format, such as an MP4 format. Then, the process goes to Step **S20**.

(56) The control unit **26** multiplexes the sound and the image (Step **S20**). Specifically, the multiplexing unit **46** multiplexes the image that has been subjected to the compression process at Step **S18** and the corresponding sound that has been subjected to the compression process at Step **S12**, and generates a single piece of compressed data. More specifically, the multiplexing unit **46** packs each piece of the compressed data of the image and the sound in a predetermined size. The multiplexing unit **46** performs a multiplexing process by the MPEG multiplexing method in which a timestamp of the same time is assigned to packs that need to be reproduced at the same time, on the basis of a synchronous timer signal of a counter (not illustrated) at 90 kHz, 27 MHz, or the like, for example. Then, the process goes to Step **S22**.

(57) The control unit **26** acquires the biological information on the user U (Step **S22**). Specifically, the biological information acquisition unit **34** acquires the biological information, such as a fingerprint, on the user U from the biological sensor **16**. Then, the process goes to Step **S24**.

(58) The control unit **26** determines whether the user U is authenticated (Step **S24**). Specifically, the authentication unit **36** compares the biological information on the user U that is acquired at Step **S22** and the biological information **24B** that is stored in the storage unit **24**, and if the pieces of the biological information match with each other, the authentication unit **36** determines that the user U is authenticated. If it is determined that the user U is authenticated (Step **S24**; Yes), the process goes to Step **S26**. If it is determined that the user U is not authenticated (Step **S24**; No), the process goes to Step **S42**.

(59) If it is determined as Yes at Step **S24**, the control unit **26** generates an encryption key (Step **S26**). Specifically, the encryption unit **48** generates an encryption key that is unique to the user U on the basis of the biological information on the user U such that personal authentication is available without management. In other words, in the present embodiment, the encryption key is generated from the biological information that is not realized even by the subject person, so that information on the key itself is not stolen by a third party. Then, the process goes to Step **S28**.

(60) The control unit **26** encrypts the compressed data (Step **S28**). Specifically, the encryption unit **48** encrypts the compressed data that has been multiplexed at Step **S20**, by several-fold and a unit, for example. For the encryption, an encryption method called Data Encryption Standard (DES) or Advanced Encryption Standard (AES) is used. DES and AES are algorithms for data encryption using a common key cryptographic method. DES is a block cipher that divides data into blocks of 64 bits and collectively encrypts each block. DES has a key length of 56 bits and is handled as 64

bits by adding 8 bits for parity check. AES is the same as DES in that the common key cryptographic method is adopted. A key length of AES is selectable from 128 bits, 192 bits, and 256 bits, each of which is longer than the key length of DES. Therefore, AES is safer than DES. In the present embodiment, the encryption unit **48** encrypts the compressed data by using AES with 256 bits. Then, the process goes to Step **S30**.

(61) The control unit **26** stores the encrypted compressed data (Step **S30**). Specifically, the storage control unit **50** stores the compressed data that has been encrypted at Step **S28** in the storage unit **24**. Then, the process goes to Step **S42**.

(62) If it is determined as No at Step **S16**, the control unit **26** recognizes a specific target that is included in the peripheral image (Step **S32**). Specifically, the image recognition unit **42** recognizes privacy information on privacy, such as a face of a person, a license plate of a vehicle, or a house nameplate, that is included in the peripheral image by using the learning model **24A**. The image recognition unit **42** may recognize different privacy information. At Step **S32**, the image recognition unit **42** may recognize copyright information on a copyrighted material that is included in the peripheral image by using the learning model **24A**, for example. Then, the process goes to Step **S34**.

(63) The control unit **26** performs an editing process on the recognized specific target (Step **S34**). Specifically, the image editing unit **44** performs editing such that the privacy information that has been recognized from the peripheral image is not recognizable. For example, the image editing unit **44** performs image processing for reducing identify of a region that includes the privacy information such that the privacy information is not identifiable. The image processing for reducing the identity is, for example, mosaic processing. The mosaic processing is processing for replacing a predetermined image region with data of an average value. The image editing unit **44** may perform various kinds of filtering processes including a Gaussian filter or a median filter on the region including the privacy information, for example. The image editing unit **44** may perform certain conversion, such as a tone change process or a brightness change process, which includes a change of a target resolution or a specific hue, on the region including the privacy information, for example. The image editing unit **44** may perform a blurring process, a point clouding process, or a process of deleting a certain part, for example. Then, the process goes to Step **S36**.

(64) The control unit **26** performs an image compression process on the peripheral image that has been subjected to the editing process (Step **S36**). Specifically, the image processing unit **40** compresses the edited peripheral image by codec of an arbitrary system, such as MPEG-4 or H.264, and converts the image into an arbitrary file format, such as an MP4 format. Then, the process goes to Step **S38**.

(65) The control unit **26** multiplexes the sound and the image (Step **S38**). Specifically, the multiplexing unit **46** multiplexes the image that has been subjected to the compression process at Step **S36** and the corresponding sound that has been subjected to the compression process at Step **S12**, and generates a single piece of compressed data. More specifically, the multiplexing unit **46** packs each piece of the compressed data of the image and the sound in a predetermined size. The multiplexing unit **46** performs a multiplexing process by the MPEG multiplexing method in which a timestamp of the same time is assigned to packs that need to be reproduced at the same time, on the basis of a synchronous timer signal of a counter (not illustrated) at 90 kHz, 27 MHz, or the like, for example. Then, the process goes to Step **S40**.

(66) The control unit **26** stores the encrypted compressed data (Step **S40**). Specifically, the storage control unit **50** stores the compressed data that has been compressed at Step **S38** in the storage unit **24**. Then, the process goes to Step **S42**.

(67) The control unit **26** determines whether the process is to be terminated (Step **S42**). Specifically, the control unit **26** determines that the process is to be terminated if operation indicating termination of the process is received or operation of turning off a power supply is received. If it is determined that the process is to be terminated (Step **S42**; Yes), the process in FIG.

4 is terminated. If it is determined that the process is not to be terminated (Step **S42**; No), the process goes to Step **S10**.

(68) As described above, in the first embodiment, the image in the line-of-sight direction of the user **U** is stored in the storage unit **24** inside the storage apparatus **10** by using an encryption key that is generated based on the biological information on the user **U**. With this configuration, in the first embodiment, the privacy information including the fact that the user **U** has paid attention and a target in the line-of-sight direction of the user **U** are stored in an encrypted manner, so that it is possible to appropriately store the image while protecting privacies of both of the user and the target.

(69) Furthermore, in the first embodiment, as for the peripheral image of the user **U**, the privacy information included in the peripheral image is recognized, the privacy information is subjected to image processing, and the image is stored in the storage unit **24** inside the storage apparatus **10** such that the privacy information included in the peripheral image is indistinguishable. With this configuration, in the first embodiment, it is possible to store the peripheral image of the user **U** as a behavior history of the user **U**, and appropriately store the image while protecting the privacy information included in the peripheral image.

(70) Moreover, in the first embodiment, only the image in the line-of-sight direction of the user **U** is encrypted, so that it is possible to prevent an increase in a buffer memory in a storage process and prevent delay of the storage process. With this configuration, in the first embodiment, it is possible to reproduce the stored line-of-sight direction image and the stored peripheral image at a high speed while protecting the privacy information.

Second Embodiment

(71) A second embodiment will be described below. A configuration of the storage apparatus according to the second embodiment is the same as the configuration of the storage apparatus **10** illustrated in FIG. **3**, and therefore, explanation thereof will be omitted.

(72) The storage apparatus **10** according to the second embodiment is different from the first embodiment in that the image in the line-of-sight direction of the user **U** and the peripheral image are temporally synchronized with each other. Further, the second embodiment is different from the first embodiment in that the peripheral image is stored in an external apparatus that is different from the storage apparatus **10**.

(73) Process Performed by Storage Apparatus

(74) A process performed by the storage apparatus according to the second embodiment will be described below with reference to FIG. **5**. FIG. **5** is a flowchart illustrating an example of the flow of a process performed by the storage apparatus according to the second embodiment.

(75) Processes from Step **S50** to Step **S54** are the same as the processes from Step **S10** to Step **S14** illustrated in FIG. **4**, and therefore, explanation thereof will be omitted.

(76) The control unit **26** temporally synchronizes the line-of-sight direction image and the peripheral image (Step **S56**). Specifically, the image acquisition unit **32** adds timestamps to the line-of-sight direction image and the peripheral image such that the line-of-sight direction image and the peripheral image that are captured at the same time can be distinguished from each other. Then, the process goes to Step **S58**.

(77) Processes from Step **S58** to Step **S80** are the same as the processes from Step **S16** to Step **S38** illustrated in FIG. **4**, and therefore, explanation thereof will be omitted.

(78) The control unit **26** stores the encrypted compressed data in an external apparatus (Step **S82**). Specifically, the storage control unit **50** stores the compressed data that has been compressed at Step **S80** in the external apparatus that is different from the storage apparatus **10** via the communication unit **22**. Then, the process goes to Step **S84**.

(79) A process at Step **S84** is the same as the process at Step **S42** illustrated in FIG. **4**, and therefore, explanation thereof will be omitted.

(80) As described above, in the second embodiment, timestamps are added to the line-of-sight

direction image and the peripheral direction image, and each of the images is stored. With this configuration, in the second embodiment, a correspondence relationship between the line-of-sight direction image and the peripheral direction image is clarified, so that it is possible to more appropriately store the images.

(81) Furthermore, in the second embodiment, the peripheral image is stored in an external apparatus that is different from the storage apparatus **10**. With this configuration, in the second embodiment, security of the peripheral image is further increased, so that it is possible to more appropriately store the image.

Third Embodiment

(82) A third embodiment will be described below. In the third embodiment, in a storage apparatus that captures images and stores the images in chronological order while the user U is moving to different places, a process of resetting an encryption chain for block cipher is performed every time the user moves a predetermined distance.

(83) A configuration of a storage apparatus according to the third embodiment will be described with reference to FIG. 6. FIG. 6 is a block diagram illustrating a configuration example of the storage apparatus according to the third embodiment.

(84) As illustrated in FIG. 6, a storage apparatus **10a** is different from the storage apparatus **10** illustrated in FIG. 3 in that a storage unit **24a** stores therein map information **24C**. The storage apparatus **10a** is different from the storage apparatus **10** illustrated in FIG. 3 in that a global navigation satellite system (GNSS) receiving unit **28** is provided. The storage apparatus **10a** is different from the storage apparatus **10** illustrated in FIG. 3 in that a control unit **26a** includes a location information acquisition unit **56**. Meanwhile, it is assumed that the storage unit **24a** stores therein the map information **24C**, but embodiments are not limited to this example. It is sufficient for the storage unit **24a** to acquire map information **24c** via the communication unit **22** and store therein the map information **24C**. In other words, the storage unit **24a** need not always store therein the map information **24C** in advance, but it is sufficient to store the map information **24C** on an as-needed basis.

(85) The GNSS receiving unit **28** is a device that detects location information on the storage apparatus **10a** (the user U). The location information in this example is earth coordinates. In the present embodiment, the GNSS receiving unit **28** is what is called a GNSS module that receive radio waves from satellites and detects the location information on the storage apparatus **10a**.

(86) The map information **24C** is data including location information on an existing building or a natural object, and is data including earth coordinates and the location information on the existing building or the natural object.

(87) The location information acquisition unit **56** acquires the location information from the GNSS receiving unit **28**. The location information acquisition unit **56** determines whether the storage apparatus **10a** (the user U) is located in a predetermined area on the basis of the location information that is acquired by the GNSS receiving unit **28** and map information **30B**. In the present embodiment, it is preferable that precision of the location information is within about 100 meters (m).

(88) In the second embodiment, an encryption unit **48a** encrypts an image that is captured around the user, by block cipher. The block cipher is a method of dividing an image into predetermined blocks and then performing encryption. The block cipher includes two systems as rules for handling block data other than the encryption process. One of the systems is an electric codebook (ECB) mode, and the other one of the systems is an output feedback (OFB) mode. In the present embodiment, the OFB mode is used.

(89) Before explanation of the OFB mode, the ECB mode will be described with reference to FIG. 7A and FIG. 7B. FIG. 7A is a diagram for explaining encryption in the ECB mode. FIG. 7B is a diagram for explaining decryption in the ECB mode.

(90) As illustrated in FIG. 7A, in the ECB mode, plaintext block data **100** is encrypted by using a

predetermined encryption key, and ciphertext block data **102** is obtained. To decrypt the ciphertext block data **102**, as illustrated in FIG. 7B, the ciphertext block data **102** is encrypted by using the same encryption key as the encryption key that is used for encrypting the plaintext block data **100**, and the plaintext block data **100** is obtained. In the ECB mode, it is possible to obtain the same result if the plaintext block data **100** is the same data. In the ECB mode, the same encryption is adopted for all pieces of block data; therefore, the method is simple, easy, easily implementable, and speedy, so that it is possible to randomly access any block data in encrypted data and start decryption. However, the ECB mode is disadvantageous in that a pattern may be analyzed by a third party by comparison between the plaintext block data **100** and the ciphertext block data **102**, the ciphertext may be fraudulently decrypted, and data may be falsified by replacing data in units of block data.

(91) The OFB mode will be described with reference to FIG. 8A and FIG. 8B. FIG. 8A is a diagram for explaining encryption in the OFB mode. FIG. 8B is a diagram for explaining decryption in the OFB mode.

(92) With reference to FIG. 8A, a method of encrypting three pieces of plaintext block data, that is, first plaintext block data **120**, second plaintext block data **122**, and third plaintext block data **124**, will be described. As illustrated in FIG. 8A, in the OFB mode, an initial vector **110** is encrypted by a predetermined encryption key, and a first encrypted vector **112** is generated. By calculating exclusive-OR between the first plaintext block data **120** and the first encrypted vector **112**, first ciphertext block data **130** is obtained. In the OFB mode, to encrypt the second plaintext block data **122**, the first encrypted vector **112** is further encrypted by a predetermined encryption key, and a second encrypted vector **114** is generated. By calculating exclusive-OR between the second plaintext block data **122** and the second encrypted vector **114**, second ciphertext block data **132** is obtained. In the OFB mode, to encrypt the third plaintext block data **124**, the second encrypted vector **114** is further encrypted by a predetermined encryption key, and a third encrypted vector **116** is generated. By calculating exclusive-OR between the third plaintext block data **124** and the third encrypted vector **116**, third ciphertext block data **134** is obtained. The initial vector **110**, the first encrypted vector **112**, the second encrypted vector **114**, and the third encrypted vector **116** are called as a key stream. The initial vector **110**, the first encrypted vector **112**, the second encrypted vector **114**, and the third encrypted vector **116** are also called as an encryption chain.

(93) A method of decrypting the three pieces of ciphertext block data, that is, the first ciphertext block data **130**, the second ciphertext block data **132**, and the third ciphertext block data **134**, will be described with reference to FIG. 8B. As illustrated in FIG. 8B, in the OFB mode, the initial vector **110** is encrypted by a predetermined encryption key, and the first encrypted vector **112** is generated. By calculating exclusive-OR between the first ciphertext block data **130** and the first encrypted vector **112**, the first plaintext block data **120** is obtained. In the OFB mode, to decrypt the second ciphertext block data **132**, the first encrypted vector **112** is further encrypted by a predetermined encryption key, and the second encrypted vector **114** is generated. By calculating exclusive-OR between the second ciphertext block data **132** and the second encrypted vector **114**, the second plaintext block data **122** is obtained. In the OFB mode, to decrypt the third ciphertext block data **134**, the second encrypted vector **114** is further encrypted by a predetermined encryption key, and the third encrypted vector **116** is generated. By calculating exclusive-OR between the third ciphertext block data **134** and the third encrypted vector **116**, the third plaintext block data **124** is obtained. The initial vector **110**, the first encrypted vector **112**, the second encrypted vector **114**, and the third encrypted vector **116** are called as a key stream. The initial vector **110**, the first encrypted vector **112**, the second encrypted vector **114**, and the third encrypted vector **116** are also called as an encryption chain.

(94) Process Performed by Storage Apparatus

(95) A process performed by the storage apparatus according to the third embodiment will be described with reference to FIG. 9. FIG. 9 is a flowchart illustrating the flow of the process

performed by the storage apparatus according to the third embodiment.

(96) The control unit **26a** acquires current location information on the storage apparatus **10a** (the user U) (Step **S90**). Specifically, the location information acquisition unit **56** acquires the current location information on the storage apparatus **10a** (the user U) on the basis of the location information that is acquired by the GNSS receiving unit **28** and the map information **30B**. Then, the process goes to Step **S92**.

(97) Processes from Step **S92** to Step **S100** are the same as the processes from Step **S10** to Step **S18** illustrated in FIG. **4**, and therefore, explanation thereof will be omitted.

(98) The control unit **26a** multiplexes the sound and the image (Step **S102**). Specifically, a multiplexing unit **46a** multiplexes the image that has been subjected to the compression process at Step **S18** and the corresponding sound that has been subjected to the compression process at Step **S12**. Further, the location information acquired at Step **S90** is added, and a single piece of compressed data is generated. More specifically, the multiplexing unit **46a** packs the compressed data of the image and the sound, which are associated with the location information, in a predetermined size. The multiplexing unit **46a** performs a multiplexing process by the MPEG multiplexing method in which a timestamp of the same time is assigned to packs that need to be reproduced at the same time, on the basis of a synchronous timer signal of a counter (not illustrated) at 90 kHz, 27 MHz, or the like, for example. Then, the process goes to Step **S104**.

(99) Processes at Step **S104** and Step **S106** are the same as the processes at Step **S22** and Step **S24** illustrated in FIG. **4**, and therefore, explanation thereof will be omitted.

(100) If it is determined as Yes at Step **S106**, the control unit **26a** generates an encryption key (Step **S108**). Specifically, the encryption unit **48a** generates a key stream of encryption keys of respective encryption blocks that are included in an encryption chain, on the basis of the initial vector that is prepared in advance. Then, the process goes to Step **S110**.

(101) The control unit **26a** encrypts the compressed data (Step **S110**). Specifically, the encryption unit **48a** calculates exclusive-OR between the compressed data and the key stream by using the key stream that is generated at Step **S106**, and encrypts the compressed data by an encryption chain in which a plurality of pieces of encryption block data are continued. Then, the process goes to Step **S112**.

(102) A process at Step **S112** is the same as the process at Step **S30** illustrated in FIG. **3**, and therefore, explanation thereof will be omitted.

(103) The control unit **26a** determines whether the user has moved a predetermined distance or more (Step **S114**). Specifically, the location information acquisition unit **56** determines whether the storage apparatus **10a** (the user U) has moved the predetermined distance or more on the basis of the location information that is acquired by the GNSS receiving unit **28** and the map information **24C**.

(104) FIG. **10** is a diagram for determining whether the storage apparatus **10a** (the user U) has moved the predetermined distance or more. FIG. **10** illustrates a trajectory of the storage apparatus **10a** (the user U) on a map. It is assumed that the storage apparatus **10a** moves from a point **P1** to a point **P5**. The storage apparatus **10a** captures images in the line-of-sight direction of the user U, encrypts the captured images, and stores the encrypted images while the user U is visiting tourist spots between the point **P1** and the point **P5**.

(105) The location information acquisition unit **56** acquires the location information on the storage apparatus **10a** (the user U) while the storage apparatus **10a** (the user U) is moving from the point **P1** to the point **P5**. In this case, for example, the map information **24C** includes area information on an area **A1**, an area **A2**, an area **A3**, an area **A4**, and an area **A5**. In other words, the map information **24C** is associated with the area **A1** to the area **A5** in advance. The location information acquisition unit **56** is able to determine an area in which the storage apparatus **10a** (the user U) is located, on the basis of the location information that is acquired by the GNSS receiving unit **28** and the map information **24C**.

(106) The location information acquisition unit **56** determines whether the storage apparatus **10a** (the user U) has moved between the areas, for example. The location information acquisition unit **56** determines whether the storage apparatus **10a** (the user U) has moved from the point **P1** in the area **A1** to the point **P2** in the area **A2**, for example. The location information acquisition unit **56** determines whether the storage apparatus **10a** (the user U) has moved from the point **P2** in the area **A2** to the point **P3** in the area **A3**, for example. The location information acquisition unit **56** determines whether the storage apparatus **10a** (the user U) has moved from the point **P3** in the area **A3** to the point **P4** in the area **A4**, for example. The location information acquisition unit **56** determines whether the storage apparatus **10a** (the user U) has moved from the point **P4** in the area **A4** to the point **P5** in the area **A5**, for example. In the present embodiment, the location information acquisition unit **56** determines that the storage apparatus **10a** (the user U) has moved the predetermined distance or more if, for example, the storage apparatus **10a** (the user U) has moved to a different area.

(107) If it is determined that the storage apparatus **10a** (the user U) has moved the predetermined distance or more (Step **S114**; Yes), the process goes to Step **S116**. If it is determined that the storage apparatus **10a** (the user U) has not moved the predetermined distance or more (Step **S114**; No), the process goes to Step **S128**.

(108) If it is determined as Yes at Step **S114**, the control unit **26a** resets the encryption chain (Step **S116**). Specifically, the encryption unit **48a** resets a value of the initial vector for generating the encryption key, in accordance with a predetermined rule. In other words, in the present embodiment, for example, every time the storage apparatus **10a** (the user U) moves to a different area, the initial vector is reset and an encrypted image is separately stored in the storage apparatus **10a** for each area. Then, the process goes to Step **S128**.

(109) Processes from Step **S118** to Step **S128** are the same as the processes from Step **S32** to Step **S42** illustrated in FIG. **4**, and therefore, explanation thereof will be omitted.

(110) As described above, in the third embodiment, the initial vector for generating an encryption key is reset if the storage apparatus **10a** (the user U) has moved the predetermined distance or more. With this configuration, in the third embodiment, it is possible to increase security and more appropriately store the image.

(111) Furthermore, in the third embodiment, an image is separately stored for each predetermined distance, for example, for each area. With this configuration, in the third embodiment, when reproducing an image of a specific area, it is not needed to decrypt all of images, so that it is possible to reproduce the image of the specific area.

Fourth Embodiment

(112) A fourth embodiment will be described below. FIG. **11** is a block diagram illustrating a configuration example of a storage apparatus according to the fourth embodiment. A storage apparatus **10b** according to the fourth embodiment is different from the storage apparatus **10a** illustrated in FIG. **6** in that an encryption unit **48b** of a control unit **26b** has a different function.

(113) In the fourth embodiment, when the location information acquisition unit **56** determines that the storage apparatus **10b** (the user U) has moved a predetermined distance or more, the encryption unit **48b** resets the initial vector and generates an initial vector and an encryption key based on location information on a moved location.

(114) Process Performed by Storage Apparatus

(115) A process performed by the storage apparatus according to the fourth embodiment will be described with reference to FIG. **12**. FIG. **12** is a flowchart illustrating an example of the flow of a process performed by the storage apparatus according to the fourth embodiment.

(116) Processes from Step **S130** to Step **S156** are the same as the processes from Step **S90** to Step **S116** illustrated in FIG. **9**, and therefore, explanation thereof will be omitted.

(117) The control unit **26b** generates an encryption key on the basis of the current location information on the storage apparatus **10b** (the user U) (Step **S158**). Specifically, in the present

embodiment, the AES encryption method is used, a block cipher has a size of 128 bits, an encryption key has a size of 256 bits, and an initial vector has a value of 128 bits. It is preferable that the encryption unit **48b** generates the initial vector with a random value to make decryption of the encryption difficult. In the present embodiment, the encryption unit **48b** generates the value of the initial vector on the basis of the current location information. For example, the encryption unit **48b** may generate the initial vector on the basis of the latitude and the longitude of the current location. The encryption unit **48b** may generate the encryption key on the basis of the current location information on the storage apparatus **10b** (the user U). The encryption unit **48b** may generate the encryption key on the basis of the current location information on the storage apparatus **10b** (the user U) and the biological information on the user U. It is preferable for the encryption unit **48b** to generate the initial vector with a different value for each message to be encrypted. Then, the process goes to Step **S170**.

(118) Processes from Step **S160** to Step **S170** are the same as the processes from Step **S118** to Step **S128** illustrated in FIG. **9**, and therefore, explanation thereof will be omitted.

(119) As described above, in the fourth embodiment, the initial vector and the encryption key are generated on the basis of the current location information on the storage apparatus **10b** (the user U). With this configuration, in the fourth embodiment, it is possible to increase complexity of an algorithm for generating the initial vector and the encryption key, so that it is possible to further improve safety.

(120) Furthermore, in the fourth embodiment, the image is separately stored for each predetermined distance, for example, for each area. Moreover, in the fourth embodiment, it is possible to encrypt the image of each specific area by using the initial vector that is generated in accordance with location information on the image and by using the encryption key, so that it is possible to generate a large number of combinations of the initial vectors and the encryption keys. With this configuration, in the fourth embodiment, it is possible to further improve the safety.

(121) A program for performing the storage method described above may be provided by being stored in a non-transitory computer-readable storage medium, or may be provided via a network such as the Internet. Examples of the computer-readable storage medium include optical discs such as a digital versatile disc (DVD) and a compact disc (CD), and other types of storage devices such as a hard disk and a semiconductor memory.

(122) According to the present disclosure, it is possible to appropriately store a video in a line-of-sight direction of a user.

(123) The storage apparatus of the present embodiment is applicable to a wearable device that is worn by a user.

(124) Although the present disclosure has been described with respect to specific embodiments for a complete and clear disclosure, the appended claims are not to be thus limited but are to be construed as embodying all modifications and alternative constructions that may occur to one skilled in the art that fairly fall within the basic teaching herein set forth.

Claims

1. A storage apparatus comprising: a biological sensor configured to detect biological information on a user; an image acquisition unit configured to acquire, from an image capturing unit, an image that is captured around the user; an image processing unit configured to separate the image around the user into a line-of-sight direction image related to an image in a line-of-sight direction of the user and a peripheral image related to an image other than the line-of-sight direction image; an encryption unit configured to generate an encryption key for encrypting the line-of-sight direction image based on the biological information on the user, and to encrypt the line-of-sight direction image by using the encryption key; a storage control unit configured to store the line-of-sight direction image that is encrypted by the encryption unit in a storage unit; and a location information

- acquisition unit configured to acquire location information on the user, wherein when the location information on the user indicates movement of a predetermined distance or more, the encryption unit is configured to reset an encryption chain related to the encryption key and generate an encryption key for encryption based on the location information on the user.
2. The storage apparatus according to claim 1, further comprising: an image determination unit configured to determine whether the peripheral image includes privacy information; and an image editing unit configured to edit the privacy information such that the privacy information is indistinguishable when the peripheral image includes the privacy information.
3. The storage apparatus according to claim 1, wherein the image processing unit is configured to temporally synchronize the line-of-sight direction image and the peripheral image, and the storage control unit is configured to store the line-of-sight direction image in the storage unit and to store the peripheral image in an external apparatus via a communication unit.
4. The storage apparatus according to claim 1, wherein when it is determined that the user has moved between areas that are determined in advance based on the location information on the user and map information that is stored in the storage unit, the encryption unit is configured to reset an encryption chain related to the encryption key.
5. A storage method comprising: acquiring location information on a user; detecting biological information on the user; acquiring, from an image capturing unit, an image that is captured around the user; separating the image around the user into a line-of-sight direction image related to an image in a line-of-sight direction of the user and a peripheral image related to an image other than the line-of-sight direction image; generating an encryption key for encrypting the line-of-sight direction image based on the biological information on the user; encrypting the line-of-sight direction image by using the encryption key; and storing the encrypted line-of-sight direction image in a storage unit, wherein the generating of the encryption key includes, when the location information on the user indicates movement of a predetermined distance or more, resetting an encryption chain related to the encryption key and generating an encryption key for encryption based on the location information on the user.
6. A non-transitory computer-readable storage medium storing a program causing a computer to execute: acquiring location information on a user; detecting biological information on the user; acquiring, from an image capturing unit, an image that is captured around the user; separating the image around the user into a line-of-sight direction image related to an image in a line-of-sight direction of the user and a peripheral image related to an image other than the line-of-sight direction image; generating an encryption key for encrypting the line-of-sight direction image based on the biological information on the user; encrypting the line-of-sight direction image by using the encryption key; and storing the encrypted line-of-sight direction image in a storage unit, wherein the generating of the encryption key includes, when the location information on the user indicates movement of a predetermined distance or more, resetting an encryption chain related to the encryption key and generating an encryption key for encryption based on the location information on the user.
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